Zoho Analytics Integration

1. Overview

1.1 Description

Zoho Analytics Integration connects **Zoho Analytics** (business intelligence and reporting platform) with **Newo Platform**.

It enables:

- Verifying caller identity against a vendor database
- Looking up invoice status by load number (due date, payment status, amount due, payee)
- Verifying rate details by load number (rate, payee, payment amount)
- Identifying caller's company/location with fuzzy name matching

This integration is designed for **logistics companies** who need **an AI phone agent to answer vendor (carrier/transporter) inquiries about invoices and rates** using data stored in Zoho Analytics tables.

1.2 Use Cases

- When a vendor calls and asks about invoice status → Agent verifies identity, queries the invoices table, and reads back due date, payment status, amount due, and payee
- When a vendor calls to verify a rate → Agent queries the rates table and reads back rate, payee, and payment amount
- When a caller needs to be verified → Agent checks the vendors table by company name (and optionally phone number)
- When a caller mentions a company → Agent identifies the correct location from the locations table, handling multiple matches with disambiguation

1.3 Supported Features

Feature	Supported	Notes
Invoice Status Lookup	✅	By load number + vendor name
Rate Verification	✅	By load number
Caller Identity Verification	✅	By vendor name; phone number optional
Location Identification	✅	LIKE matching for partial company names
Multi-Region Support	✅	US, EU, IN, AU, JP, SA, CA data centers
OAuth2 Authentication	✅	Standard redirect flow with token auto-refresh
Custom Canvas Scenarios	✅	Logistics-specific scenarios and intent types
Booking/Scheduling	❌	Not applicable — this is a data query integration
Write Operations	❌	Read-only — agent never modifies Zoho Analytics data

2. Requirements

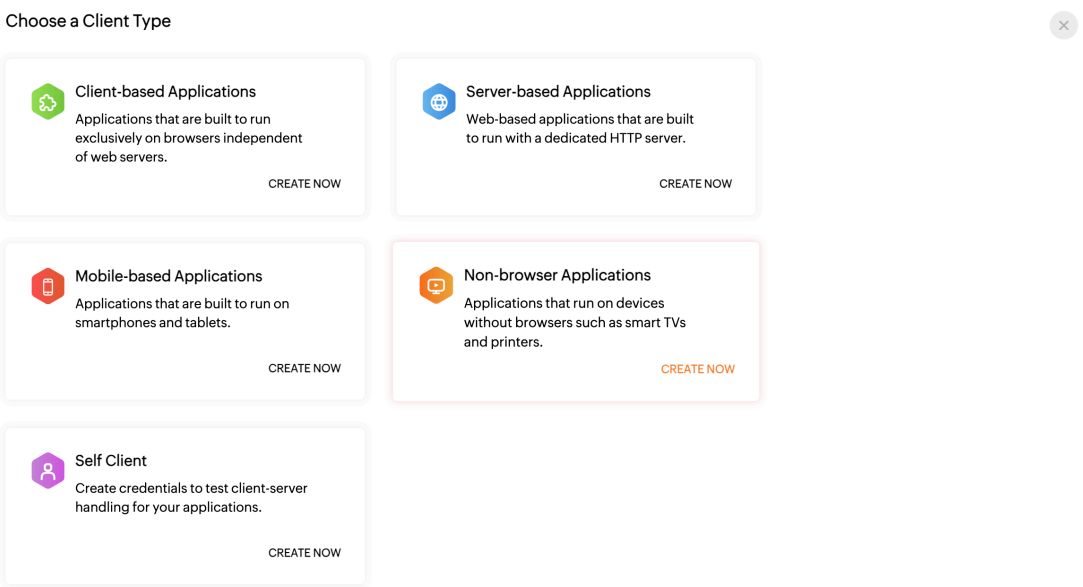
Before installation:

- Active **Zoho Analytics** account with API access (free tier works)
- **Server-based Applications** client registered in Zoho API Console
- A Zoho Analytics workspace with 4 tables: **Invoices**, **Rates**, **Vendors**, **Locations**
- Data synced from your TMS (Transportation Management System) to these tables

3. How to Setup

3.1 Step 1 — Register OAuth Client in Zoho

1. Go to Zoho API Console
2. Click **Add Client** → select **Server-based Applications**
3. Fill in:
   - **Client Name**: `Newo Zoho Analytics`
   - **Homepage URL**: `https://newo.ai`
   - **Authorized Redirect URIs**: `https://static.newo.ai/oauth/index.html`
4. Copy **Client ID** and **Client Secret**



3.2 Step 2 — Prepare Tables in Zoho Analytics

Create a workspace with 4 tables:

Invoices — invoice/payment data:

Column Name	Data Type
Load Number	Plain Text
Vendor Name	Plain Text

Due Date	Date
Payment Status	Plain Text
Amount Due	Number
Payee	Plain Text

Rates — rate and payment data:

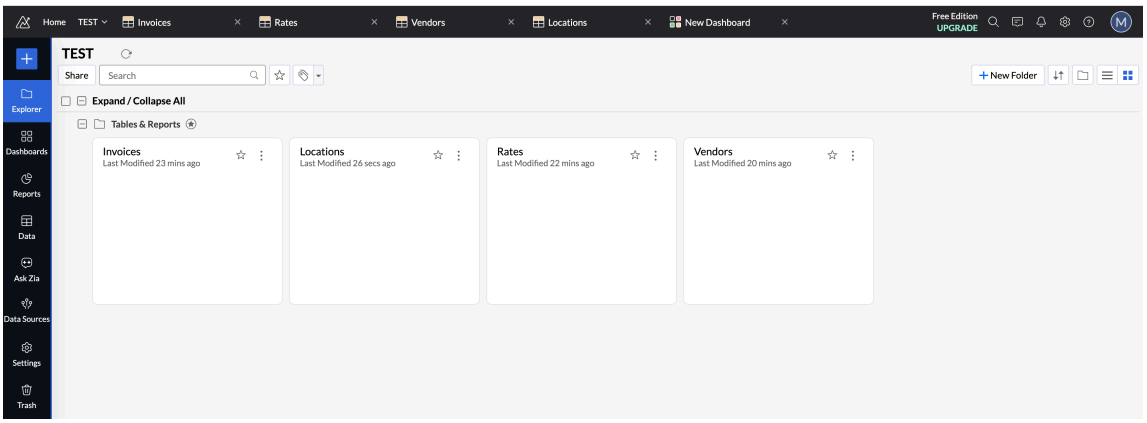
Column Name	Data Type
Load Number	Plain Text
Rate	Number
Payee	Plain Text
Payment Amount	Number

Vendors — vendor directory:

Column Name	Data Type
Vendor Name	Plain Text
Phone	Plain Text
Contact Person	Plain Text
Email	Plain Text

Locations — company/location directory:

Column Name	Data Type
Company Name	Plain Text
Location	Plain Text
Region	Plain Text
Account ID	Plain Text



> ***Important**: Column names must match exactly (case-sensitive, with spaces). The integration queries by these exact column names.*

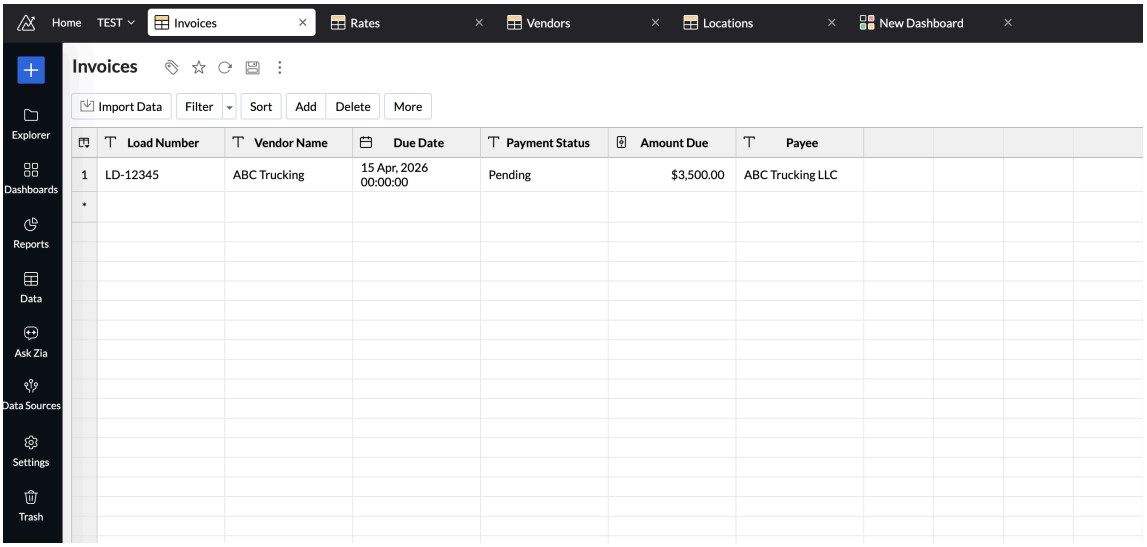
3.3 Step 3 — Get Workspace ID and View IDs

Open each table in Zoho Analytics. The View ID is visible in the browser URL:

```
https://analytics.zoho.eu/workspace/WORKSPACE_ID/view/VIEW_ID
```

Note down:

- **Workspace ID**
- **View ID** for each of the 4 tables



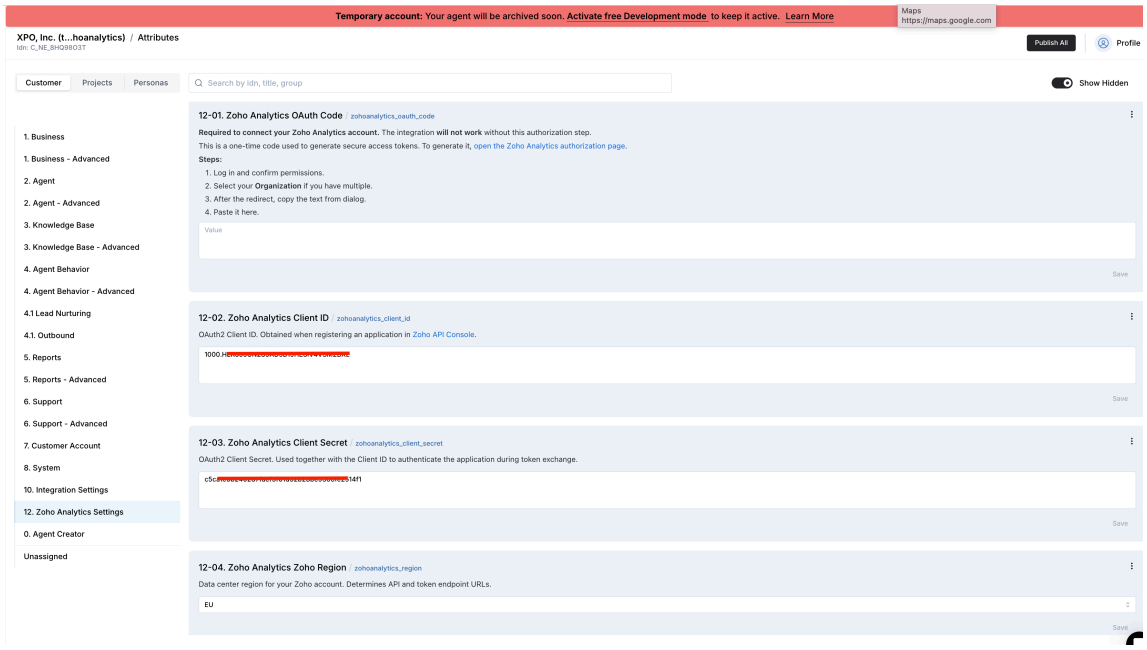
3.4 Step 4 — Get Organization ID

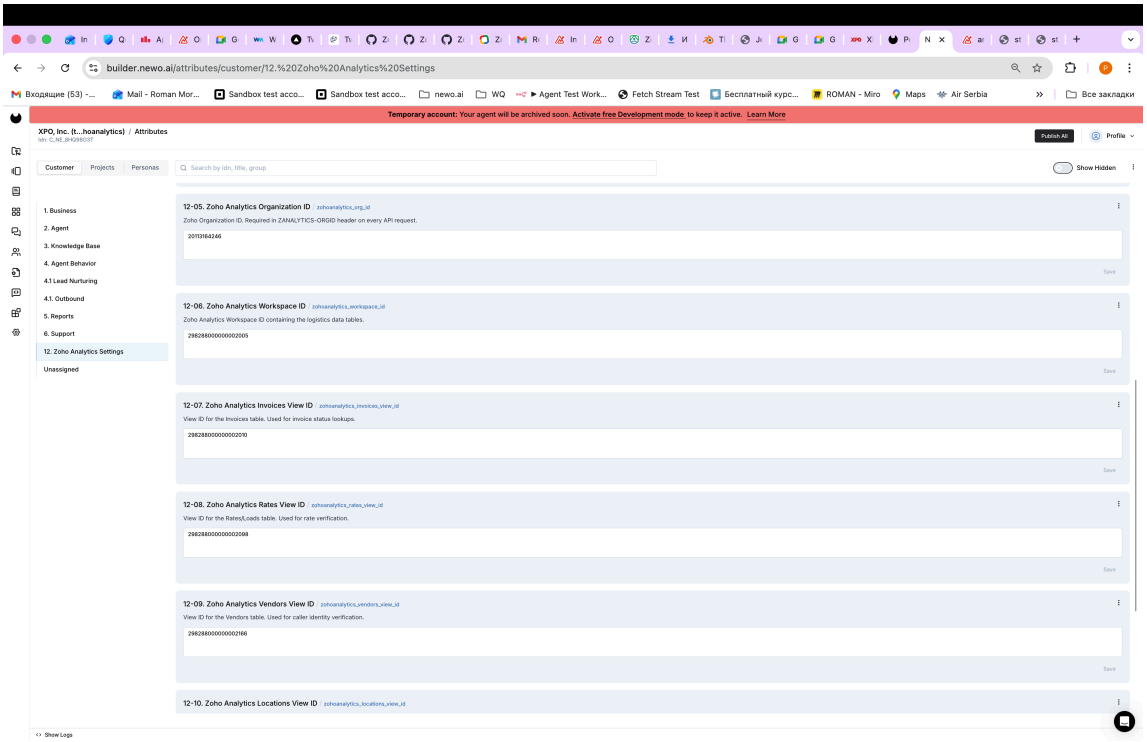
Go to **Zoho Analytics → Setup → General → Organization Details**. Copy the **Organization ID** from the URL or the settings page.

3.5 Step 5 — Connect in Newo Platform

1. Open **builder.newo.ai → Project → Attributes**
2. Set the following attributes:

Attribute	Required	Description
`zohoanalytics_client_id`	✅	OAuth2 Client ID from Zoho API Console
`zohoanalytics_client_secret`	✅	OAuth2 Client Secret from Zoho API Console
`zohoanalytics_region`	✅	Data center region: US, EU, IN, AU, JP, SA, or CA
`zohoanalytics_org_id`	✅	Zoho Organization ID
`zohoanalytics_workspace_id`	✅	Zoho Analytics Workspace ID
`zohoanalytics_invoices_view_id`	✅	View ID for the Invoices table
`zohoanalytics_rates_view_id`	✅	View ID for the Rates table
`zohoanalytics_vendors_view_id`	✅	View ID for the Vendors table
`zohoanalytics_locations_view_id`	❌	View ID for the Locations table
`zohoanalytics_setup_scenarios`	❌	Auto-add logistics scenarios to canvas (default: True)
`zohoanalytics_override_agent_attributes`	❌	Override SA attributes on publish (default: True)





3. **Fill in Client ID and Client Secret first**, then **Publish** the project
4. After publish, click the OAuth link in the **OAuth Code** attribute to authorize
5. Paste the code and **Publish** again

After publish the integration automatically:

- Connects to the right Zoho data center based on your region
- Authorizes against Zoho with the OAuth code you provided

- Enables the four logistics tools (verify caller, identify location, invoice lookup, rate verification) for the agent
- Adds the matching scenarios to the Workflow Builder Canvas so the agent knows when to use each tool

4. How to Use

4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a code-phrase the agent says, detected by the ConvoAgent tool caller
- An **Action** is a query to Zoho Analytics Sync Export API

Available Triggers:

Trigger	Tool Event	Description
Verify Caller	`zohoanalytics_verify_caller_tool`	Agent says "Let me verify your information..."
Invoice Status Lookup	`zohoanalytics_query_invoice_status_tool`	Agent says "Let me look up the invoice..."
Rate Verification	`zohoanalytics_query_rate_verification_tool`	Agent says "Let me check the rate..."
Identify Location	`zohoanalytics_identify_location_tool`	Agent says "Let me look up that company..."

What the agent does behind each trigger:

- **Verify Caller** — looks up the vendor by name (and optionally phone) in your **Vendors** table.
- **Invoice Status Lookup** — pulls the row for the requested load number + vendor from your **Invoices** table.
- **Rate Verification** — pulls the row for the requested load number from your **Rates** table.
- **Identify Location** — finds matches by company name in your **Locations** table; if more than one location matches, the agent asks the caller to clarify.

The same caller is never asked the same question twice in a single call — once a result is on screen, the agent answers from it without re-running the tool. Cleared automatically when the conversation ends.

4.2 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel) by interacting with the agent as a vendor calling about invoices or rates.

4.3 Testing and Verification

To test the integration:

1. **Setup**: Publish the project and verify attributes appear in the builder (Client ID, Region, View IDs)
2. **OAuth**: Click the authorization link in the OAuth Code attribute, authorize in Zoho, paste the code back into the attribute, and publish again. The integration stores the tokens automatically.
3. **Caller Verification**: Say "Hi, this is [vendor name from Vendors table]" → Agent should say "Let me verify your information" → Confirm verification succeeded

4. **Invoice Lookup**: Ask "What's the status of my invoice for load [load number]?" → Agent should return due date, payment status, amount, payee
5. **Rate Verification**: Ask "What's the rate for load [load number]?" → Agent should return rate, payee, payment amount

If no action occurs:

- Ensure Client ID and Client Secret are set before first Publish
- Verify Region matches your Zoho account data center (check URL: `.zoho.com` = US, `.zoho.eu` = EU, etc.)
- Confirm all View IDs are filled (check for empty values)
- Re-publish after any attribute changes
- Check that table column names match exactly: `Load Number`, `Vendor Name`, `Due Date`, `Payment Status`, `Amount Due`, `Payee`, `Rate`, `Payment Amount`, `Phone`, `Contact Person`, `Email`, `Company Name`, `Location`, `Region`, `Account ID`
- Verify data exists in Zoho Analytics tables (empty tables return "not found")

> *Note: This integration performs **read-only** queries against Zoho Analytics. No data is modified in Zoho Analytics through the agent.*

5. FAQ

Q: Does the integration modify data in Zoho Analytics? A: No. The integration is read-only — it only looks up rows. No data is created, updated, or deleted.

Q: How does data get into Zoho Analytics tables? A: Data must be synced from your TMS (Transportation Management System) separately. Zoho Analytics supports imports from databases, CSV, cloud storage, and 250+ integrations. Data freshness depends on your sync frequency.

Q: What happens if the vendor is not found during verification? A: The agent informs the caller that verification failed and offers to transfer to a human representative.

Q: Can the agent handle multiple invoice lookups in one call? A: Yes. After answering an invoice question, the agent asks "Is there anything else?" The caller can ask about another load number.

Q: What if the table has more than 1 million rows? A: The current setup is designed for tables up to 1 million rows. If you expect more, contact us — a different lookup approach will be needed.

Q: Which Zoho regions are supported? A: US, EU, India, Australia, Japan, Saudi Arabia, and Canada. Set the correct region in the **Region** attribute.

Q: Can I use Zoho Analytics free tier? A: Yes. The free tier (2 users, 5 workspaces, 10,000 rows) is sufficient for testing and small deployments.